

Radiotherapy for laryngeal (larynx) cancer

Radiotherapy uses high-energy rays to destroy laryngeal cancer cells.

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What is radiotherapy?

Radiotherapy uses high-energy rays to destroy the cancer cells, while doing as little harm as possible to normal cells.

How and when radiotherapy is used to treat [laryngeal cancer](#) depends on the [stage of the cancer](#).

We understand that having treatment can be a difficult time for people. We're here to support you. If you want to talk, you can:

- Call the Macmillan Support Line on [0808 808 00 00](#).
- [Chat to our specialists online](#).
- [Visit our laryngeal \(larynx\) cancer forum](#) to talk with people who have been affected by laryngeal cancer, share your experience, and ask an expert your questions.

Radiotherapy for early-stage laryngeal cancer

Radiotherapy is often the first choice of treatment for [early stage](#) cancer of the larynx. Your cancer doctor may advise you to have [radiotherapy](#) or [endoscopic surgery](#). Both treatments help you keep your voicebox.

The aim of radiotherapy for early cancer of the larynx is to cure the cancer. This is called radical radiotherapy.

Radiotherapy for locally advanced laryngeal cancer

If the cancer is larger, or spreads to [lymph nodes](#) or tissues nearby it is called ([locally advanced](#)) In this case, [radiotherapy](#) may be combined with other treatments.

You may have radiotherapy:

- after surgery to destroy any remaining cancer cells, and reduce the risk of it coming back
- after surgery, combined with chemotherapy ([chemoradiation](#))
- instead of surgery, combined with chemotherapy ([chemoradiation](#))
- combined with a [targeted therapy](#) drug.

If you have surgery first, you usually start chemoradiation or radiotherapy 6 or 7 weeks later.

Radiotherapy for advanced laryngeal cancer

Sometimes it is not possible to cure the cancer. In this case, the main aim of radiotherapy is to help reduce symptoms. This is sometimes called palliative radiotherapy.

You may have palliative radiotherapy to:

- help control the cancer for a period of time
- shrink a tumour that is causing problems with swallowing or breathing
- help improve symptoms, if the cancer has spread to other areas of the body
- reduce symptoms – this might happen if the cancer has spread to [lymph nodes](#) and these are causing problems such as pain, bleeding or breaking through the skin.

How radiotherapy is given

Radiotherapy is usually given from outside the body. This is called [external beam radiotherapy](#).

High-energy x-rays are directed at the cancer from a radiotherapy machine. Radiotherapy is not painful, but you do have to lie still for a few minutes while the treatment is being given.

You usually have external beam radiotherapy as an outpatient in the hospital radiotherapy department. It is normally given as a number of short treatments. These are called treatment sessions or fractions.

You may have radiotherapy:

- Monday to Friday, with a rest at the weekend (this is the most common method)
- more than once a day (hyperfractionation).

Treatment may take 3 to 7 weeks. This is called a course of treatment. Your cancer doctor, specialist nurse or radiographer will discuss the treatment with you.

If you are having radiotherapy to help with symptoms, you may only need a short course or a single treatment session.

Intensity-modulated radiotherapy (IMRT)

IMRT is the main type of external beam radiotherapy used to treat cancer of the larynx. It shapes the radiotherapy beams and allows the radiographer to give different doses of radiotherapy to different parts of the treatment area.

This means the healthy tissue surrounding the tumour has lower doses of radiotherapy. This can help reduce the risk of [side effects](#) and [late effects](#). It may also allow the radiographer to give higher doses of radiotherapy to the tumour.

Research has shown that having IMRT may reduce some long-term side effects, including having a dry mouth. This is because radiotherapy given in this way avoids the salivary glands, where saliva (spit) is produced.

Some people may have regular scans during their treatment. This is to make sure the radiotherapy is targeting the treatment area. Sometimes the treatment may need adjusting, for example, if you lose weight during treatment.

Volumetric modulated arc therapy (VMAT)

VMAT is a type of IMRT that is used to treat many different cancers. It is sometimes used to treat larynx cancer.

VMAT delivers a continuous dose of radiation as the machine rotates 360 degrees around the body. This treatment accurately shapes the radiation dose to the tumour while minimising the dose to the healthy tissue surrounding it.

Giving the radiotherapy in this way makes it very accurate and shortens the treatment time.

Conformal radiotherapy (CRT)

Many types of external beam radiotherapy are conformal. This means the beams are specially shaped to fit the treatment area. It may be used to give lower doses of radiotherapy.

Planning your radiotherapy

To make sure your radiotherapy is as effective as possible, it must be carefully planned. Planning makes sure the radiotherapy is aimed directly at the cancer. This means it causes the least possible damage to surrounding healthy tissues.

Your cancer doctor plans your treatment with a radiotherapy team who are experts in giving you radiotherapy.

Radiotherapy planning is usually done during a visit to the radiotherapy department. Sometimes you may need to make more than one visit.

Planning can take up to 2 hours. Your team will let you know how long it is likely to take and answer any questions you may have.

Radiotherapy masks

It is important that you lie still and in the correct position during each treatment. To help you stay still, you wear a plastic mesh mask for each session of radiotherapy. The radiographer, or mask room technician, will explain how the mask is made.

The mask is designed so you can see and breathe normally while wearing it. It is sometimes called a mould, head shell or cast.

We have more information about [masks for radiotherapy](#).

Planning a CT scan

After your mask is made, you have a [CT scan](#) of the area to be treated. The radiographers take measurements to plan the treatment specifically for you. Some people also have an [MRI scan](#) as part of their radiotherapy planning.

The radiographer enters the measurements and the information from the scans into the radiotherapy planning computer. Your doctors then use this to plan your treatment precisely.

Radiotherapy treatment sessions

At the beginning of each session, the radiographers will make sure you are in the correct position. They will tell you how long your treatment will take. They then carefully fit you with your mask and make sure you are comfortable.

When everything is ready, they leave the room and the treatment starts. The treatment itself is not painful. It does not make you radioactive. You will not be a risk to anyone during your treatment.

The radiographers will watch you on cameras. If you need to talk to them during your treatment, you can use the intercom.

In some treatment rooms, you can listen to music to help you relax during treatment. Ask your radiographers if this is possible.

During treatment, the radiotherapy machine may stop and move into a new position. This is so you can have radiotherapy from different directions. Some radiotherapy machines make a small beeping noise to tell you when the treatment is starting and when it has finished.

About our information

References



Below is a sample of the sources used in our laryngeal cancer information. If you would like more information about the sources we use, please contact us at cancerinformationteam@macmillan.org.uk

ESMO Annals of Oncology. Squamous cell carcinoma of the oral cavity, larynx, oropharynx and hypopharynx. EHNS-ESMO-ESTRO Clinical Practice Guidelines for diagnosis, treatment and follow up. 2020. Available from <https://www.esmo.org/guidelines/ehns-esmo-estro-clinical-practice-guideline-squamous-cell-carcinoma-of-the-head-and-neck> (accessed Jan 2022).

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Reviewers



This information has been written, revised and edited by Macmillan Cancer Support's Cancer Information Development team. It has been reviewed by expert medical and health professionals and people living with cancer. It has been approved by Senior Medical Editor, [Dr Claire Paterson](#), Consultant Clinical Oncologist.

Our cancer information has been awarded the [PIF TICK](#). Created by the Patient Information Forum, this quality mark shows we meet PIF's 10 criteria for trustworthy health information.

Date reviewed

Reviewed: 01 June 2022 | Next review: 01 June 2025

ⓘ This content is currently being reviewed. New information will be coming soon.

Our cancer information meets the [PIF TICK](#) quality mark.

This means it is easy to use, up-to-date and based on the latest evidence. Learn more about [how we produce our information](#).

How we can help

Macmillan Support Line

The Macmillan Support Line is a free and confidential service for people living with and affected by cancer. If you need to talk, we'll listen.



Call us free on 0808 808 0000

8am to 8pm, 7 days a week. You can talk to our cancer specialists about anything.



Email the Macmillan Support Line

Email the Macmillan Support Team if you have any questions.

Chat online

Available 8am to 7.45pm, 7 days a week. For confidential, real-time support if you don't want to call.



Online Community

Chat online anonymously to others who understand what you are going through. Our community is available 24/7 and has dedicated forums where you can get advice and ask our experts.



Find cancer support services in your area

What's going on near you? Find out about support groups, where to get information and how to get involved with Macmillan where you live.

Get help